

MANSTON, United Kingdom

WMO: 037970

Lat: 51.3464N

Lon: 1.3356E

Elev: 164

StdP: 14.61

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 35047

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.4	30.9	21.7	16.4	31.7	24.5	18.7	32.8	36.9	48.1	33.1	47.9	9.5	60	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.7	78.1	65.7	74.8	64.4	72.1	63.0	67.2	75.1	65.7	72.4	64.2	70.0	11.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	90.7	70.1	63.0	86.6	68.6	61.8	83.0	67.3	31.7	75.1	30.5	72.6	29.4	70.2	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.0	25.5	22.8	DB	24.6	85.2	2.4	3.3	22.9	87.6	21.5	89.6	20.2	91.4	18.5	93.8	(4)
(5)			WB	23.6	70.6	2.5	1.3	21.8	71.6	20.3	72.3	18.9	73.1	17.0	74.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	41.1	41.5	44.4	48.7	53.9	59.3	63.6	64.0	59.7	53.8	46.9	42.5	(6)
(7)		DBStd	9.53	5.34	5.31	4.91	5.02	4.75	4.49	4.17	3.98	4.06	4.82	5.10	5.63	(7)
(8)		HDD50	1178	278	241	183	82	16	0	0	0	0	20	118	240	(8)
(9)		HDD65	4953	740	659	640	490	344	181	77	67	166	348	543	698	(9)
(10)		CDD50	1784	2	2	8	42	137	278	422	434	291	137	25	6	(10)
(11)		CDD65	87	0	0	0	0	1	9	34	35	7	1	0	0	(11)
(12)		CDH74	362	0	0	0	0	4	45	145	144	21	3	0	0	(12)
(13)		CDH80	45	0	0	0	0	0	4	19	21	1	0	0	0	(13)
(14)	Wind	WSAvg	11.7	13.4	13.1	12.3	11.3	11.1	10.4	10.3	10.5	10.8	11.8	12.1	12.8	(14)
(15)	Precipitation	PrecAvg	22.6	1.8	1.3	1.5	1.7	1.6	1.8	1.8	2.3	2.6	2.6	2.0		(15)
(16)		PrecMax	34.6	4.5	4.9	3.7	5.0	4.3	4.1	4.8	7.0	10.0	9.8	5.5	4.1	(16)
(17)		PrecMin	15.9	0.4	0.2	0.0	0.2	0.0	0.1	0.3	0.2	0.1	0.3	0.7	0.4	(17)
(18)		PrecStd	3.9	0.9	0.9	0.9	0.9	0.8	1.0	1.0	1.3	1.9	1.9	1.3	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	56.3	60.2	69.1	73.3	79.0	82.6	82.6	77.1	69.6	60.4	56.3	(19)
(20)			MCWB	52.1	50.8	53.0	58.0	62.1	65.9	67.3	66.9	64.0	62.5	57.8	53.9	(20)
(21)		2%	DB	52.9	53.0	56.4	62.9	68.6	74.0	77.8	78.0	71.7	64.7	58.1	54.6	(21)
(22)			MCWB	50.6	49.8	50.8	54.6	59.5	64.0	65.8	65.5	63.3	60.0	55.6	52.6	(22)
(23)		5%	DB	51.1	51.1	54.0	59.3	65.1	70.4	74.7	74.4	68.7	62.5	56.5	53.0	(23)
(24)			MCWB	49.0	48.0	49.4	52.5	57.4	61.8	64.3	64.3	61.4	58.7	54.0	51.1	(24)
(25)		10%	DB	49.3	49.3	51.8	56.6	62.3	67.6	71.8	71.5	66.3	60.8	54.6	51.3	(25)
(26)			MCWB	47.2	46.7	48.1	51.1	55.6	60.1	62.9	62.7	60.3	57.6	52.3	49.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	52.3	54.0	58.9	63.5	67.9	69.0	69.2	66.7	63.2	58.5	54.7	(27)
(28)			MCD B	54.0	54.9	58.3	66.9	70.6	76.9	79.0	79.2	72.9	67.7	59.7	55.7	(28)
(29)		2%	WB	51.1	50.4	51.8	55.5	60.5	65.1	67.2	67.1	64.6	61.1	56.3	53.0	(29)
(30)			MCD B	52.5	52.4	55.1	61.2	66.8	72.1	75.5	74.5	69.6	63.7	57.7	54.3	(30)
(31)		5%	WB	49.3	48.6	50.3	53.5	58.5	63.0	65.5	65.5	63.0	59.7	54.5	51.3	(31)
(32)			MCD B	50.9	50.5	53.2	58.1	63.7	68.9	72.6	71.9	67.0	61.9	56.1	52.8	(32)
(33)		10%	WB	47.4	47.1	48.6	51.8	56.5	61.0	63.9	64.0	61.5	58.3	52.5	49.5	(33)
(34)			MCD B	49.1	49.0	51.3	55.6	61.1	66.3	69.8	69.6	65.1	60.3	54.3	51.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	8.8	10.5	12.2	12.6	13.4	13.9	13.7	11.9	10.3	8.7	7.9	(35)
(36)			MCDBR	9.6	10.5	13.4	16.2	17.5	17.8	18.3	17.6	14.9	11.7	9.5	8.4	(36)
(37)			MCWBR	9.2	8.1	9.1	9.7	10.1	9.9	8.5	7.9	8.1	7.9	8.2	8.2	(37)
(38)		5% WB	MCDBR	9.7	9.9	12.5	15.1	16.1	16.5	16.6	15.8	13.5	10.8	9.1	8.4	(38)
(39)			MCWBR	9.6	8.5	9.1	9.6	10.0	9.9	8.4	7.9	8.1	8.2	8.6	8.4	(39)
(40)		Clear-Sky Solar Irradiance	taub	0.332	0.348	0.385	0.403	0.411	0.412	0.409	0.401	0.381	0.365	0.339	0.323	(40)
(41)	taud		2.392	2.395	2.292	2.276	2.306	2.326	2.341	2.385	2.428	2.442	2.425	2.383	(41)	
(42)	Ebn at Noon		215	245	255	265	268	268	267	263	255	236	211	198	(42)	
(43)	Edh at Noon		20	25	34	38	39	38	37	34	29	24	19	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	251	466	827	1330	1580	1755	1669	1391	1041	594	300	193	(44)	
(45)		RadStd	23	69	102	122	112	138	144	104	79	56	35	25	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (73 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page